CaseStudy Tech

Implementation Plan

CSCI 5520 - AI-Driven Software Development

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Stable IDs (US-##, RC-##, DD-##, OQ-##, ASM-##, RISK-##) refer to the Software Design Document v1.0 (2026-10-04). This plan sequences the work. GitHub Project #1 is the authoritative work queue for detailed tasks, status and order (Section 7).

1 Implementation Overview

1.1 Implementation Objective

Deliver the CaseStudy Tech MVP student practice loop defined in SDD §2.2:

- Google sign-in (US-01).
- One of three vetted scenarios with an informed start (US-02).
- A time-bounded voice-first session with text fallback (US-03, US-04).
- Safe recovery and conduct safeguards (US-05, US-06).
- A separate post-session grading pass producing an evidence-linked developmental report (US-07).
- Local report export (US-08).
- A scores-only history (US-09).

The deliverable is a working React Native build for web, iOS (TestFlight/internal only, RC-15) and Android. It runs on the managed Google Cloud services in SDD §5.

Scope boundaries for every phase:

- Requirements RC-01 to RC-22 are hard constraints. When a technical choice conflicts with one, the requirement wins.
- The SDD §2.3 non-goals stay out. That means no instructor or institutional tools, no content authoring, no live coaching or live scores, no stored or replayable sessions, no model training or self-hosting, no clinical or emergency use, no enterprise or billing features, and no additional disciplines or localization.

1.2 Source Design Artifacts

Artifact	Location	Authority
Software Design Document v1.0 (2026-10-04)	`output/CSCI_5520_CaseStudy_Tech_Software_Design_Document_Final.docx`	Authoritative for scope, stories, requirements, decisions
Interactive UI/UX prototype (Figma Make)	https://www.figma.com/make/liPGx5kd9Cb6LeVcHHgddH/Interactive-UI-Mockup-for-CaseStudy-Tech	Authoritative for screens and flow; not production code (RISK-05)
Architecture diagram — SDD Figure 1	`docs/02-gcp-architecture.png` (Lucidchart source)	Authoritative for components and data flow
Experience flow — SDD Figure 2	SDD §6.1	Authoritative for the student journey and recovery branches
Screen figures — SDD Figures 3–10	SDD §6.2	Visual references for Issues
Discovery notes (2026-09-16)	`docs/01-discovery-notes.md`	Background only; superseded by the SDD
Account boundaries decision	`docs/decisions/0001-account-boundaries.md` (draft PR #62)	Authoritative for sign-in, token handling, Firestore ownership and deletion
Existing backlog	GitHub Project #1: 10 epics and 40 child Issues created from the SDD before Lab 3	Reconciled in place by §5.1

1.3 Planning Assumptions & Scope Guardrails

These assumptions and decisions change the build order. Each was decided by the project owner during Lab 3 planning unless it is marked **open**.

ID	Assumption / guardrail	Effect on sequencing	Source
PA-1	Solo builder, roughly six-week MVP window. If the schedule slips, cut polish, never data-retention or safety requirements.	Phases are ordered so each one leaves a demonstrable loop.	RISK-06
PA-2	**Speech provider is chosen first.** A provider evaluation spike compares voice quality, round-trip latency, cost per session and verified zero-retention terms before session work starts. The provider stays behind a replaceable interface. **Latency target:** voice round trip (end of student speech to first persona audio) at p50 ≤ 2.0 s and p95 ≤ 3.5 s. A provider that misses it is rejected, unless the owner records an exception under §6 CD-1.	The provider decision sits on the critical path ahead of voice session work. Voice-first per DD-08, with text always available.	OQ-01, RC-05, RC-16, RISK-02
PA-3	**Synthetic data until zero-retention is confirmed.** Development and testing use only synthetic transcripts and test accounts. No real student session runs until zero-retention and no-training terms are confirmed in writing for every model and speech API in use.	Persona and grading integration can proceed; written confirmation is a release gate.	OQ-02, RC-05, DD-04
PA-4	**Rubric anchors are hybrid and built from free sources.** The six categories and weights stay fixed (RC-17). The grader scores against observable behaviors for each 1-to-5 level (WHO EQUIP style); students see developmental level labels. Anchor text is original and cites only free instruments: EQUIP/ENACT, FIS, CTS-R, Bogo OSCE, CSWE EPAS and SIRI-2. Ivey microskills and MI principles appear only as cited concepts. CCS-R and the MITI manual are not used. Survey: `docs/lab3/rubric-survey.md`.	An early grading-phase Issue drafts the anchor spec.	OQ-03 (resolved), RC-17, RC-18
PA-10	**Calibration replaces faculty review.** *This supersedes SDD OQ-03 and the faculty-review requirement in RISK-03.* No faculty review gates the anchors or the crisis scenario. Instead, release requires (a) the grader to score a labeled calibration set of synthetic transcripts (strong, weak and harmful per category, plus crisis critical-failure cases) with every category within ±1 level of its label on at least 90% of transcripts, and with every crisis critical-failure case triggering the safety override; and (b) the conduct and crisis safeguards to pass every labeled test example (see §3.4). Calibration transcripts are drafted by a model family different from the grader's, and the owner approves every label before grader work starts (§6 CD-5).	Calibration-set creation is on the grading phase's critical path.	RISK-01, RISK-03, RISK-04, RC-19, RC-20
PA-5	**Session time limit: 10 minutes for all three scenarios.** The limit lives in per-scenario configuration, so changing it later is a config edit.	Session timer work must read the limit from scenario config.	OQ-04 (resolved)

ID	Assumption / guardrail	Effect on sequencing	Source
PA-6	**The report PDF is generated on the device.** After the report is delivered, the narrative never goes back to the server, so RC-03 holds by design. Web, iOS and Android each need a working PDF path.	The export deliverable sits in Phase 5. No server render endpoint exists.	OQ-05 (resolved), RC-03, US-08
PA-7	**Expo + FastAPI.** The client is Expo (React Native with react-native-web), giving one codebase for web, iOS and Android. The session and grading services are Python FastAPI on Cloud Run. The SDD fixes the managed GCP services but not these two choices.	The scaffold is the first Phase 1 deliverable. The stack is recorded in `docs/ARCHITECTURE.md`.	DD-06, DD-09, RC-14
PA-8	The prototype is a design reference, not code to copy. Its known defects (empty-state history, score-ring percent sign) are fixed during implementation.	UI Issues link figures, not prototype markup.	RISK-05
PA-9	Users are adult postsecondary students with Google accounts. A FERPA review is needed before institutional sales but does not block the MVP.	No MVP phase.	ASM-01, ASM-02
PA-11	**The plan follows the code.** Work already built in draft PR #62 stays where it is: the accounts service, Terraform, Firestore rules and account deletion are Phase 1. The existing Issues are reconciled in place, not recreated (§5).	Phase 1 is partly in progress. Account deletion moves from Phase 6 to Phase 1.	Owner decision, 2026-10-30
PA-12	**Invite-only access.** A Google sign-in creates an account that reaches nothing until the owner approves it from an admin queue. Each request emails the owner. Approval is enforced in every API and in the Firestore rules. This adds an admin role and an access record (name and Google email), both recorded exceptions to the SDD's single role and RC-04's "scores only" (`docs/POLICY.md` ACC-4 to ACC-9, DAT-2). Modelled on the system-design-course invite flow.	Five Phase 1 Issues (#72 to #76) sit between the identity guard and the library. The email needs OD-3 first.	Owner decision, 2026-10-31 (POLICY interview)

2 Phase Summary

Six phases, each ending in a demonstrable increment. Phase 1 is partly in progress in draft PR #62 (PA-11). Phase 2 runs alongside Phase 1. Phase 5's anchor spec and calibration set are authored from day one; its grader code starts after Phase 1 on synthetic transcripts and integrates with real transcripts once Phase 3 lands (§4).

Phase	Goal / Increment	Primary SDD Sources	Key Deliverables	Depends On	Candidate Issues
1	Foundation & access — an approved student signs in, browses the three scenarios, reviews preparation info and can delete their account	US-01, US-02, RC-01, RC-02, RC-04, RC-06, RC-13, RC-14, RC-22, DD-05, DD-06, DD-09, PA-12	RN shell (web/iOS/Android), accounts service with Google sign-in, owner approval gate with admin queue, Terraform GCP baseline, Gateway + Cloud Run, Firestore owner rules, account deletion, Secret Manager, content-free logging, scenario config incl. per-scenario time limit	None	10 — §3.1

Phase	Goal / Increment	Primary SDD Sources	Key Deliverables	Depends On	Candidate Issues
2	Provider & compliance spike — speech vendor chosen and zero-retention confirmed in writing	OQ-01, OQ-02, RC-05, RC-16, RISK-02, DD-04	Provider comparison + decision record, Vertex zero-retention settings, replaceable speech interface	None (parallel to 1)	5 — §3.2
3	Practice session — timed voice-first session with text, persona replies, End session, offline check	US-03, US-04, RC-03, RC-07, RC-10, RC-12, DD-01, DD-08	Session service, persona prompts, voice + text UI, timer, transient state	1; 2 for voice	6 — §3.3
4	Safety & recovery — conduct safeguards, recovery paths, crisis disclosures that do not end the session	US-05, US-06, RC-08, RC-19, RC-20, RISK-04	Conduct safeguard, recovery flows, labeled safeguard test set	3	6 — §3.4
5	Grading & report — evidence-linked report with crisis override, passing the calibration gate	US-07, US-08, RC-09, RC-10, RC-17, RC-18, RC-19, OQ-05, RISK-01	Anchor spec, separate grader, citation validation, calibration set, report UI, on-device export	1 (synthetic); 3 for integration	7 — §3.5
6	History & release — scores-only history, release gates met	US-09, RC-04, RC-11, RC-12, RC-14, RC-15, PA-3, PA-10	Firestore score record and history, accessibility audit, platform parity, TestFlight build, release gate review	4, 5	6 — §3.6

3 Detailed Implementation Phases

Figure numbers refer to SDD §6. "Prototype state" names a state in the Figma Make reviewer panel. Every phase uses synthetic data only until the PA-3 release gate is met.

3.1 Phase 1 — Foundation & access

Goal

A student signs in with Google on web, iOS or Android. Once the owner approves the account (PA-12), the student browses the three vetted scenarios and opens a scenario's preparation screen. The start controls follow the opt-in and connectivity rules. A student can sign out or request account deletion. Starting a session is wired up in Phase 3.

Status (2026-10-30): partly in progress. Draft PR #62 (`feature/4-access-accounts`) carries the Expo app, the accounts service, the Terraform baseline, the Firestore rules, account deletion and 58 passing tests. It still needs real OAuth clients, iOS evidence from a Mac, a fix for the #61 dependency findings, and alert destinations plus abuse controls for the unauthenticated token-exchange routes. The plan follows that code (PA-11).

Source Requirements / Design References

US-01, US-02 · RC-01, RC-02, RC-04, RC-06, RC-07 (pre-session connectivity check), RC-13, RC-14, RC-21, RC-22 · DD-05, DD-06, DD-07, DD-09 · PA-5, PA-7, PA-11, PA-12 · `docs/POLICY.md` ACC-4 to ACC-9 · `docs/decisions/0001-account-boundaries.md` · Architecture: client, Identity Platform, API Gateway, Cloud Run, Firestore, Secret Manager, Cloud Logging (Fig. 1).

Deliverables

- An Expo project (React Native + react-native-web) at `apps/student` that produces web, iOS and Android builds. CI runs lint, typecheck, tests and a dependency audit.
- The GCP baseline in Terraform (`infra/`): Cloud Run, API Gateway generated from a template, Identity Platform, Firestore and Secret Manager.
- The accounts service (`services/accounts`), the first FastAPI service on Cloud Run behind API Gateway. The gateway validates the token. The service then re-verifies the original Firebase token, its revocation status and the Google provider. Later services reuse this pattern.
- Google sign-in, sign-out and a sign-in failure state. The server exchanges the Google token through Identity Platform, so no API key ships in the client. ID tokens live only in memory.
- Firestore security rules with emulator tests: a student reads only their own collection, and every client write is denied.
- Invite-only access (PA-12): an access record per account, an approval check in every protected API and rule, an owner email per request, the admin queue and the student's access-state screens.
- Account deletion on the Account screen. It needs confirmation and first writes a denial marker, so access fails closed even during an outage. Cleanup retries until the account and its data are gone, within 30 days.
- Provider credentials served from Secret Manager. A CI check scans the client bundle for secrets.
- Accessible UI primitives and the Figma visual tokens, so every screen starts accessible and on brand (RC-11, PA-8).
- A structured-logging helper that drops session-content fields, with a test that proves it.
- Read-only scenario configuration for the three scenarios: objective, format, time limit, feedback timing, content notice, crisis opt-in flag and a release flag. Each scenario has its own time limit, set to 10 minutes (PA-5). The release flag hides a scenario held back under §6 CD-3.
- Welcome, Library and Preparation screens. They include the "What this is not" educational boundary, the crisis opt-in checkbox and an offline check that disables both start buttons.

Phase Acceptance Criteria

- On all three builds, a valid Google sign-in from an approved account opens the library and sign-out returns to Welcome. No other sign-in method appears (US-01, RC-01).
- An unapproved account reaches only its access state, sign-out and account deletion, in the APIs and the rules as well as the UI. A revoke applies on the next request (PA-12).
- An unauthenticated client cannot reach the library. An unauthenticated API request gets a 401 (RC-02).
- Rules tests deny cross-user reads and all client writes (RC-02).
- Deletion requires confirmation and blocks access as soon as it is confirmed. The account and its data are removed within 30 days, and an interrupted cleanup resumes (RC-04).
- The library shows the three vetted scenarios, minus any whose release flag is off, and offers no create or edit control (RC-22).
- Each preparation screen shows the objective, format, configured time limit and content notice. On the crisis scenario, both start buttons stay disabled until the student checks the opt-in. Every start button is disabled while the device is offline (US-02, RC-13, RC-07).
- CI fails when a credential pattern appears in the client bundle. The logging test shows that content fields are never written (RC-06, RC-03).

Dependencies & Sequence Notes

No phase dependencies; Phase 2 runs alongside. Human setup is required first: the GCP project, billing, the Identity Platform Google provider and OAuth clients for each platform, and an Apple Developer account for TestFlight (RC-15). The scaffold must merge before any other Phase 1 Issue. Changing the PA-5 time limit later is a config edit, not a code change. PR #62 bundles several Issues because it was already written. From here on, every PR closes out one Issue (§5).

Relevant UI/UX Artifacts

Fig. 1 (architecture), Fig. 2 (flow from Welcome to Preparation, and the Account path), Fig. 3 (welcome and library), Fig. 4 (crisis preparation and opt-in), Fig. 10 (native layouts with bottom tabs). Prototype states: sign-in failure, offline.

Candidate GitHub Issues

- Scaffold the Expo app with web, iOS and Android builds and CI
- Provision the GCP baseline in Terraform
- Sign students in and out with Google through the accounts service
- Verify identity at the gateway and the service, and lock Firestore to the signed-in owner
- Let students delete their account with confirmation and fail-closed cleanup
- Require owner approval before an account reaches the app, with an admin queue and an owner email (added 2026-10-31, PA-12: #72 to #76)
- Keep secrets server-side and clear the dependency audit findings
- Keep session content out of logs with a content-free logging helper
- Define the configuration for the three vetted scenarios, including per-scenario time limits
- Build the accessible primitives and Figma visual tokens
- Build the library and preparation screens with the crisis opt-in and offline check

3.2 Phase 2 — Provider & compliance spike

Goal

A speech provider is chosen on measured evidence. Zero-retention and no-training terms are confirmed in writing for every hosted speech and model API in use. If a confirmation is missing, real-student use stays blocked and the record says so. All session code reaches speech through a replaceable interface.

Source Requirements / Design References

OQ-01, OQ-02 · RC-05, RC-06, RC-16 · DD-01, DD-04, DD-08, DD-09 · RISK-02 · PA-2, PA-3 · Architecture: speech APIs and Vertex AI persona and scoring inference (Fig. 1).

Deliverables

- A speech provider interface in the session service (speech-to-text and text-to-speech, or speech-to-speech) with a fake implementation for tests.
- Prototype adapters for OpenAI and Google Cloud Speech behind that interface.
- A benchmark harness that drives scripted synthetic turns and records round-trip latency, voice quality ratings and cost per 10-minute session.
- A provider decision record (ADR) covering latency p50/p95, quality, cost, availability and data-use terms, with the decision and the rejected option.

- A Vertex AI data-governance record listing models, region, zero-retention settings and abuse-monitoring logging status. The record points to where the written confirmations are kept. Signed vendor documents stay out of the repo.

Phase Acceptance Criteria

- The chosen provider meets p50 ≤ 2.0 s and p95 ≤ 3.5 s round-trip latency, measured over at least 30 scripted synthetic turns (PA-2, RISK-02).
- Written zero-retention and no-training terms are on file for the chosen speech provider and each Vertex AI model. For any that are missing, the record marks real-student use as blocked (RC-05, PA-3).
- A contract test passes against both the fake and the chosen adapter. Swapping providers changes only adapter code (RC-16).
- Provider credentials load from Secret Manager and do not appear in the client bundle (RC-06).

Dependencies & Sequence Notes

The research can start on day one. The adapters plug into the Phase 1 service skeleton. Vendor confirmations are external and slow, so request them at the start of the phase. Missing confirmations do not block Phases 3–5 on synthetic data, but they do block release (§3.6). Voice work in Phase 3 cannot start until the decision record merges.

Relevant UI/UX Artifacts

Fig. 1: speech API and Vertex AI boxes and the zero-retention note. This phase has no screens.

Candidate GitHub Issues

- Define the replaceable speech provider interface and its fake
- Benchmark OpenAI and Google speech for latency, quality and cost
- Obtain written zero-retention and no-training terms for speech and Vertex AI
- Configure Vertex AI for zero data retention and record the settings
- Record the speech provider decision as an ADR

3.3 Phase 3 — Practice session

Goal

A student runs a timed session against a vetted persona, by voice or by text. The student can end the session, or the timer ends it. The complete transcript is handed off for grading. No live scores appear, and no session content persists.

Source Requirements / Design References

US-03, US-04 · RC-03, RC-07, RC-10, RC-12, RC-13 · DD-01, DD-02, DD-08 · PA-3, PA-5 · Architecture: session service, Vertex AI persona inference, speech APIs, temporary conversation state (dashed path, Fig. 1).

Deliverables

- Session endpoints for start, turn and end. The server enforces the time limit from scenario config, and session state lives only in memory, discarded on end or abandon.
- A persona prompt harness with three vetted persona prompts on Vertex AI. In the crisis persona, a disclosure is a turn and never a session end.
- A text session screen with a remaining-time display and an End session control.

- A voice check for connection and microphone readiness, and a voice session screen with text fallback always visible. If the microphone is denied, the student continues in text.
- A memory-only turn store on the client, which the Phase 4 recovery flows read.
- A completed-session handoff contract to the grading service (complete transcript plus completion status). The session service does no grading.

Phase Acceptance Criteria

- Either mode can start on a connected build. Text mode never requests the microphone. Voice mode does not start after the microphone is denied and offers text instead (US-03).
- The persona stays in scenario. The session ends on End session and at the configured limit, and the server rejects turns after either end condition (US-04).
- No score, hint or correction appears during a session (RC-10).
- After a session ends, neither Firestore nor the logs contain audio or transcript content, and the service holds no session state. Automated tests cover this (RC-03).
- Session start is blocked offline, and no screen claims offline support (RC-07).
- A text session can be completed with VoiceOver and TalkBack (RC-12).

Dependencies & Sequence Notes

The text path depends only on Phase 1. The voice path also needs the Phase 2 decision record. The handoff contract must be fixed early because Phase 5 builds against it.

Relevant UI/UX Artifacts

Fig. 2 (format choice, voice check, end conditions), Fig. 5 (voice and text sessions), Fig. 10 (iOS voice session). Prototype states: offline, microphone denied.

Candidate GitHub Issues

- Run sessions with a server-enforced time limit and transient state
- Drive the three vetted personas through a prompt harness on Vertex AI
- Build the text session screen with timer and End session
- Build the voice check and voice session with text fallback
- Hand completed transcripts to the grading service
- Prove no session content persists after a session ends

3.4 Phase 4 — Safety & recovery

Goal

Sessions fail safely and enforce conduct rules:

- A dropped voice session continues in text when the conversation is intact.
- Lost content or a conduct violation ends the attempt unscored.
- Crisis disclosures never end the crisis scenario.

The safeguards pass the labeled test set.

Source Requirements / Design References

US-05, US-06 · RC-08, RC-09, RC-19, RC-20 · DD-08 · RISK-04 · PA-10(b) · Architecture: conduct safeguards in the session service (Fig. 1).

Deliverables

- Recovery flows. A voice interruption with the conversation intact continues the same session in text. Lost content ends the attempt unscored and offers a new text session from the beginning. Neither flow stores the partial session.
- A conduct safeguard in the session service with three rules:
  - Immediate termination for high-confidence targeted threats, slurs or sexual harassment.
  - One warning, then termination, for repeated demeaning language.
  - Further evaluation instead of automatic termination for quoted case material, nondirected frustration and low-confidence speech recognition.
- A synthetic labeled safeguard test set with must-terminate, must-warn and must-not-terminate cases. The must-not-terminate cases include crisis disclosures.
- Screens for the conduct warning, conduct ended, voice interrupted and conversation lost states. Each says whether the conversation is safe and whether a score was saved.

Phase Acceptance Criteria

The safeguard must pass every labeled case (US-06, RC-20, RC-19, PA-10(b)):

- Every must-terminate case ends the session without a score.
- Every repeated-demeaning case gets exactly one warning before termination.
- No must-not-terminate case causes an automatic termination.

Recovery and failure handling:

- With the conversation intact, a voice interruption continues the same session in text. Lost content ends the attempt unscored, creates no history entry and offers a new text session (US-05, RC-08).
- No failure or termination path sends a transcript to grading or writes a score (RC-09).

Dependencies & Sequence Notes

This phase depends on Phase 3: the session service, the turn store and the voice path. The labeled test set can be written while Phase 3 is in progress. Passing it is a release gate, checked in §3.6.

Relevant UI/UX Artifacts

Fig. 2 (four recovery branches), Fig. 9 (recovery, conduct and failure states). Prototype states: voice interrupted, conversation lost, conduct warning, conduct ended.

Candidate GitHub Issues

- Continue a dropped voice session in text when the conversation is intact
- End lost-content attempts unscored and offer a fresh text session
- Enforce conduct rules with one warning and immediate termination
- Build the labeled conduct and crisis safeguard test set
- Keep crisis disclosures from ending the crisis scenario
- Build the recovery and conduct state screens

3.5 Phase 5 — Grading & report

Goal

A completed session produces a validated, evidence-linked developmental report. The student sees it immediately and can export it to the device. The grader passes the calibration gate.

Source Requirements / Design References

US-07, US-08 · RC-03, RC-09, RC-10, RC-17, RC-18, RC-19, RC-21 · DD-02, DD-03, DD-07 · RISK-01, RISK-03, RISK-05 · PA-4, PA-6, PA-10(a) · `docs/lab3/rubric-survey.md` · Architecture: grading and report service, Vertex AI rubric scoring (Fig. 1).

Deliverables

- A rubric anchor spec. It gives behavioral anchors for each category at each 1-to-5 level, plus the developmental labels students see, the MI principles for the resistant-client scenario and the crisis critical-failure criteria. It cites free instruments only (PA-4).
- A grading service: a separate Cloud Run FastAPI service with its own Vertex AI rubric prompt. It accepts only complete transcripts. Code computes the overall score from the RC-17 weights; the model does not.
- Evidence validation. Every cited turn must exist in the transcript and support its claim. A failed check yields no score and the grading-failed state.
- The crisis safety override: a critical safety failure becomes the primary outcome, and the average is shown as a secondary measure.
- A synthetic calibration set with labels: strong, weak and harmful transcripts per category and scenario, plus crisis critical-failure cases. A calibration runner reports agreement with the labels.
- Report screens for the standard and crisis reports. They include the developmental disclaimer and the 988 Lifeline on the crisis report, with the score-ring percent sign fixed (RISK-05).
- On-device PDF export on web, iOS and Android, and a leave-report confirmation that offers export first.

Phase Acceptance Criteria

- On the calibration set, every category is within ±1 level of its label on at least 90% of transcripts, and every crisis critical-failure case triggers the override (PA-10(a)).
- Every report has an overall score, six 1-to-5 ratings, strengths and improvements. A unit test shows that the overall score equals the RC-17 weighted formula (US-07, RC-17).
- Every cited example maps to an existing transcript turn. A report that fails validation shows the grading-failed state and saves nothing (RC-18, RISK-01).
- The grader rejects incomplete or failed transcripts (RC-09).
- The exported PDF matches the displayed report, and exporting sends no narrative to the server (US-08, RC-03, PA-6).
- No resistant-client report calls anything an MITI score (RC-18).

Dependencies & Sequence Notes

This chain is the longest path to release. The anchor spec and calibration set are documents and data, so they start on day one in the critical-path lane (§4). The grader code starts once the Phase 1 service skeleton merges. The calibration labels come from the anchor spec, never from grader output. Wiring the grader into live sessions needs the Phase 3 handoff contract. Phase 5 writes no scores; Phase 6 persists them.

Relevant UI/UX Artifacts

Fig. 6 (feedback report), Fig. 7 (crisis report with safety override), Fig. 8 left (leave-report confirmation), Fig. 10 (Android crisis report). Prototype state: grading failed.

Candidate GitHub Issues

- Write the rubric anchor specification from free instruments
- Build the synthetic calibration set and calibration runner

- Grade completed transcripts with fixed weights and the crisis override
- Reject reports whose cited evidence is not in the transcript
- Build the feedback and crisis report screens
- Export the report as a PDF on the device
- Confirm before leaving an unexported report

3.6 Phase 6 — History & release

Goal

Completed sessions save only scores, and a student sees only their own history. Every release gate is met before a real student uses the MVP on web, Android or iOS (TestFlight). Account deletion already ships in Phase 1 (PA-11); this phase makes sure it also covers the scores.

Source Requirements / Design References

US-09 · RC-02, RC-03, RC-04, RC-05, RC-11, RC-12, RC-14, RC-15 · DD-03, DD-05 · RISK-05 · PA-3, PA-10, PA-11 · Architecture: Firestore (Fig. 1).

Deliverables

- A Firestore score record with the overall score and six category scores, stored under the student's collection from Phase 1. Any other field must be justified against RC-04 in the schema Issue.
- A server-side score write that happens only from a validated report. Clients still cannot write. In the same transaction, the writer checks the Phase 1 deletion marker, so a deleted account never gains a new score.
- A score history screen with the empty state fixed (RISK-05).
- An accessibility audit and fixes: WCAG 2.2 AA on web, plus VoiceOver and TalkBack walkthroughs.
- A platform parity check on web, iOS and Android.
- A separate production GCP project in Terraform, with its own credentials and the same zero-retention, rules and logging settings as development (`docs/ARCHITECTURE.md` AD-5, #77).
- A release gate checklist and release builds: web, Android and an iOS TestFlight build. Every deploy needs a human's approval, and no agent triggers one (AG-9).

Phase Acceptance Criteria

- A student sees only their own scores. The Phase 1 rules tests still pass with the score record in place (US-09, RC-02).
- Firestore and the history screen hold no audio, transcript, excerpt or narrative content (RC-03).
- An unscored or failed attempt creates no history entry (RC-09).
- Deletion cleanup removes saved scores, and a score write after the deletion marker is rejected (RC-04).
- The core flow has no open WCAG 2.2 AA failures and passes VoiceOver and TalkBack walkthroughs (RC-11, RC-12).
- The core session, feedback, export and history flow works the same on all three builds. iOS ships only through TestFlight or internal distribution (RC-14, RC-15).
- Every release gate is met: PA-3 written confirmations, PA-10 calibration and safeguard passes, a provisioned production project that shares no credentials with development (AD-5), and a clean secrets scan (RC-05, RC-06).

Dependencies & Sequence Notes

Release depends on Phases 4 and 5. The score schema can start right after Phase 1, because the rules already exist. Each phase should run its own accessibility checks; Phase 6 is the final audit, not the first one. The release gate review comes last.

Relevant UI/UX Artifacts

Fig. 8 right (score history), Fig. 10 (native layouts). Prototype state: empty score history.

Candidate GitHub Issues

- Save scores-only records from validated reports, honoring the deletion marker
- Build the score history screen with a correct empty state
- Audit and fix accessibility (WCAG 2.2 AA, VoiceOver, TalkBack)
- Verify Figma visual fidelity and cross-screen navigation
- Verify cross-platform parity and produce the release builds
- Provision the production GCP project for real-student use (added 2026-10-31, AD-5: #77)
- Review the release gates before real student use

4 Cross-Phase Dependencies & Implementation Sequence

Critical path

The longest chain runs through grading. It starts with the anchor spec, then the labeled calibration set and the grader. Evidence validation and the calibration pass follow. Then come live integration with the Phase 3 handoff, the score write and the release gate review. Phase 1 foundations are on every path, but they are short. The provider spike and the session and safety work (Phases 2–4) form a second, shorter chain that joins at release. A slip in the anchor spec or calibration set moves the release date. A slip elsewhere usually does not.

One dependency sits outside the code. Written zero-retention and no-training confirmations from vendors (PA-3) are slow and nobody on the project controls them. They are requested on day one. They block release only, never development on synthetic data.

Shared prerequisites

Many later Issues build on these. Each one merges before its consumers start:

- **Expo scaffold and CI.** Every client Issue builds on it.
- **Terraform GCP baseline and the accounts service behind API Gateway, with Secret Manager.** Every server Issue and every credential depends on it. The accounts service sets the identity-check pattern that the session and grading services copy.
- **Content-free logging helper.** It must exist before any endpoint handles session content (RC-03, RC-06).
- **Scenario configuration.** The preparation screens, the session timer, the personas and the safeguard test set all read it.
- **Contracts used across phases:**
  - the speech provider interface and its fake (Phase 2 → 3);
  - the completed-session handoff (Phase 3 → 5);
  - the report schema (Phase 5 grader → report screens and export);
  - the Firestore rules and the deletion marker (Phase 1), which the Phase 6 score write must honor.

Parallel opportunities

Work runs in **two lanes** (Sequencing rule 1):

- **Lane 1, the critical path.** On day one it starts the anchor spec and then the calibration set. Grader code follows once the service skeleton merges.
- **Lane 2, everything else, in phase order.** It starts with the Phase 1 foundations and the Phase 2 spike, then moves through the Phase 3, Phase 4 and Phase 6 Issues.

Inside lane 2, these Issues are independent of each other and can be taken in any order:

- the Phase 2 benchmark and the Phase 1 screens;
- the text session and the provider ADR;
- the safeguard test set and the Phase 3 UI;
- the score schema and any Phase 3 or 4 work.

Sequencing rules for implementation agents

1. **Two lanes.** At most two implementation Issues are In Progress at once: one on the critical path and one off it. Waiting on vendors does not count against the limit (PA-1).
2. **Scaffold first.** The Expo scaffold merges before any other Phase 1 Issue. The Terraform baseline and the gateway pattern from the accounts service merge before any other server Issue.
3. **Contract before consumer.** A consumer starts only after its contract (listed above) is merged. Until the real producer exists, it builds against a fake.
4. **Voice waits for the decision record.** Voice session work starts only after the provider ADR merges. The text path does not wait (PA-2).
5. **Labels before the grader.** The calibration labels come from the anchor spec before the grader is tuned. They are never adjusted to match grader output (PA-10).
6. **Safeguard tests come first.** The labeled safeguard test set merges before the conduct safeguard implementation, and the implementation must pass every case (PA-10(b)).
7. **No score before validation.** Nothing writes a score until evidence validation and the score rules are both merged (RC-09, RC-18).
8. **Synthetic data until the gates.** No real student session runs until the release gate review passes (PA-3, PA-10).
9. **Accessibility in every phase.** Each UI Issue meets its own accessibility checks. The Phase 6 audit is the final check, not the first one (RC-11, RC-12).

4.1 Dependency / Sequence Table

Item / Capability	Must Follow	Reason	Can Run in Parallel With
Expo scaffold + CI	Human GCP/OAuth/Apple setup	Every client Issue builds on it	Anchor spec; provider research
Service skeleton: Terraform baseline + accounts service (Gateway, Secret Manager)	Scaffold CI; human GCP setup	Every server Issue and credential depends on it; in PR #62	Anchor spec; calibration set
Content-free logging helper	Service skeleton	Must exist before any endpoint handles session content (RC-03, RC-06)	Sign-in; scenario config

Item / Capability	Must Follow	Reason	Can Run in Parallel With
Google sign-in + unauthenticated rejection	Service skeleton	Library and every API are auth-gated (RC-01, RC-02)	Scenario config; speech interface
Firestore owner rules	Sign-in	Owner-only reads and no client writes; the score write builds on them (RC-02)	Scenario config
Account deletion (denial marker + cleanup)	Sign-in; Firestore rules	Access fails closed at once; data gone within 30 days (RC-04)	Library + preparation screens
Access approval gate (#72)	Firestore rules; account deletion	Every protected API and rule checks approval; the access record is deleted with the account (PA-12)	Scenario config; logging helper
Admin decisions (#73) → admin queue screen (#76)	Access gate	Only the server-configured admin decides (PA-12)	Owner email; access-state screens
Owner email (#74)	Access gate; OD-3	New data processor needs the owner's choice first (AG-3)	Admin decisions
Access-state screens (#75)	Access gate; shared presentation	The library mounts only after approval	Admin queue screen
Scenario configuration	Service skeleton	Prep screens, timer, personas and safeguard tests read it (PA-5)	Sign-in; logging helper
Library + preparation screens	Sign-in; access-state screens; scenario config	Screens need auth, approval and real config	Provider benchmark
Speech interface + fake	Service skeleton	Adapters and voice code target the interface (RC-16)	Phase 1 screens
Provider benchmark → ADR	Speech interface	Latency gate decides the provider (PA-2)	Text session; persona harness
Vendor zero-retention confirmations	Nothing — request day one	External and slow; release gate (PA-3)	Everything
Vertex AI zero-retention config	GCP project	Persona and grader inference run on the configured project	Phase 1 work
Session endpoints (timer, transient state)	Logging helper; scenario config; auth; OD-1 (resolved: ARCHITECTURE AD-1)	Server-enforced limit and no persistence (US-04, RC-03)	Calibration set
Persona prompt harness	Session endpoints; Vertex config	Personas run inside the session service	Text session screen
Text session screen	Session endpoints; prep screens	Text is the baseline and the fallback (DD-08)	Persona harness; provider ADR
Voice check + voice session	Provider ADR; text session screen	Voice needs the chosen provider and a working text fallback	Handoff contract
Completed-session handoff contract	Session endpoints	Phase 5 integration builds against it	Persona harness; voice session
No-persistence proof	Text and voice sessions	Proves RC-03 across both modes	Safeguard test set

Item / Capability	Must Follow	Reason	Can Run in Parallel With
Labeled safeguard test set	Scenario config	Tests come before the safeguard (PA-10(b))	Phase 3 UI
Conduct safeguard + crisis non-termination	Safeguard test set; session endpoints	Must pass every labeled case (RC-19, RC-20)	Recovery flows
Recovery flows (voice→text, lost content)	Voice session; client turn store	Recovery reads the turn store (US-05, RC-08)	Conduct safeguard
Recovery + conduct state screens	Recovery flows; conduct safeguard	Screens report states those flows produce	Report screens
Rubric anchor spec	Nothing — documentation	Head of the critical path (PA-4)	All Phase 1 and 2 work
Calibration set + runner	Anchor spec (labels); grader (runner execution)	Labels must come from anchors, not grader output (PA-10(a))	Phase 1 and 3 work
Grader (weights + crisis override)	Service skeleton; anchor spec	Scores against the anchors; code computes the weights (RC-17, RC-19)	Phase 3 session work
Evidence validation	Grader	Reports whose evidence fails validation are blocked (RC-18)	Report screens on a fake report
Feedback + crisis report screens	Report schema	Built against a fake report until the grader is live	Evidence validation
On-device PDF export + leave confirmation	Report screens	Export must match the displayed report (US-08, PA-6)	Score history screen
Live grading integration	Handoff contract; evidence validation	Real transcripts reach a validated grader only	Phase 4 work
Score record schema	Firestore owner rules; report schema	Holds only RC-04 fields under the owner's collection	Phase 3 and 4 work
Save scores only from validated reports	Evidence validation; score record; deletion marker	No score before validation (RC-09); no score after deletion (RC-04)	History screen
Score history screen	Score record	Reads only the student's own scores	PDF export
Accessibility audit	All Phase 3–6 UI	Final check across the core flow (RC-11, RC-12)	Parity check prep
Parity check + release builds	Accessibility audit; all features	Same core flow on all three builds (RC-14, RC-15)	Release gate checklist drafting
Production GCP project (#77)	Terraform baseline; Vertex zero-retention config	Real student data never shares a project or credentials with test data (AD-5)	Phases 3–6 work
Release gate review	Everything above, incl. vendor confirmations and the production project	Last step before real student use (PA-3, PA-10)	Nothing

5 GitHub Issue Decomposition Guidance

Project #1 already holds 10 epics and 40 child Issues, written from the SDD before Lab 3. Some of them already have branches, PR #62 and progress comments. The owner chose to **reconcile them in place** (2026-10-30).

Existing Issues keep their numbers and history. The ones that contradict this plan are rewritten, and new Issues fill only the gaps. Nothing is duplicated.

Sizing rules

1. **One Issue, one PR.** Each Issue should close in a single PR into `dev` that a reviewer can check in one sitting: roughly 400 changed lines or fewer, not counting lockfiles and generated files. PR #62 (80 files, 8 Issues) is the exception that set this rule (PA-11).
2. **One phase per Issue.** An Issue that crosses a phase boundary is split there. For example, scenario configuration is Phase 1 and persona prompts are Phase 3.
3. **Contracts are their own Issues.** The speech interface, the session handoff, the report contract and the score record each get an Issue that merges before any consumer (§4 rule 3).
4. **Test sets come before implementations.** Calibration labels and safeguard cases are separate Issues that merge first (§4 rules 5 and 6).
5. **Decisions are Issues too.** Spikes, vendor terms and the release gate review each end in a recorded decision. While an Issue waits on a vendor, it keeps its status, and a comment names the blocker (Project workflow §2).
6. **Epics only group work.** Code lands against child Issues. An epic tracks progress through its sub-issues.

What an agent-ready Issue contains

The body extends the project `CLAUDE.md` structure. Board fields (Status, Plan Phase, Stage, Priority, Area, Sequence, Human Review, Review Depth) live on the Project, not in the body (§7).

Section	Contents
Title	One outcome, verb first, in user or system terms ("Reject reports whose cited evidence is not in the transcript")
`## Description`	What and why. The plan phase and the SDD IDs it satisfies (US, RC, DD), plus the plan IDs that constrain it (PA, CD, OD, AG)
`## Implementation Requirements`	Checkable bullets: the constraints, the files and directories it touches, the contracts it consumes or produces
`## Acceptance Criteria`	`- [ ]` items, each tied to a named test or demo, drawn from the phase acceptance criteria in §3
`## Visual Context`	UI Issues only: SDD figure, prototype state and the Figma node from the #57 inventory. Backend Issues cite Fig. 1
`## Dependencies`	`Blocked by #n` for each §4.1 predecessor and any open decision (OD-1, OD-2)
`## Non-goals`	What is out of scope, and which Issue owns it instead
`## Testing`	The test levels expected (unit, emulator, integration, e2e), synthetic data only, and the labeled set it must pass where one applies
`## Guardrails`	The §6.4 AG numbers that apply

Review depth

Every implementation PR is merged by a human (AG-9). Two Project fields carry review (§7). **Human Review** records whether the owner has checked the Issue itself (Pending → Reviewed). **Review Depth** tells the owner how deep to look:

Level	Applies to	Owner does
Owner decision	ADRs, vendor terms, the anchor spec, calibration labels, the release gate review	Makes or approves the decision the Issue records
Deep review	Auth, Firestore rules, deletion, privacy, secrets, infrastructure, safeguards, grading	Reads the whole diff and runs or inspects the evidence
Standard review	Screens, shared UI, docs, tooling	Checks acceptance-criteria evidence and skims the diff

5.1 Issue Map

Plan phase → GitHub Issue. Every row is a real Issue on Project #1. The new ones are #63–#70, created 2026-10-30. **Change** is the reconciliation action: *keep*, *rewrite* (the body contradicted this plan), *re-phase* (its plan phase or sequence moved), *new* or *close*. The board fields for each change, and the NEW-n IDs used while it was proposed, are in `docs/lab3/backlog-proposal.md`.

ID	Issue Title	Phase	SDD Sources	Depends On	Change	Notes / Visual Context
#17	Scaffold React Native project with web, iOS, and Android builds	1	DD-06, RC-14, PA-7	Human setup	keep	Draft PR #71 (`feature/17-cross-platform-scaffold`); overlaps PR #62 in `apps/student`, so the two must be reconciled. Fig. 10
#51	Set up CI/CD pipeline with automated checks and approved deploys	1	RC-06, AG-9	#17	keep	Deploys are automated but approved by a human, which matches AG-9
#18	Provision Google Cloud baseline for MVP services	1	DD-09, RC-06	Human setup	keep	In Progress, PR #62 (`infra/`). Fig. 1
#23	Keep secrets server-side and enforce encrypted transport	1	RC-06	#18, #51	keep	Owns the client-bundle secret scan
#61	Resolve remaining student app dependency audit findings	1	RC-06	#17	keep	Bug; must be fixed before PR #62 merges
#25	Google Sign-In with Identity Platform and sign-out	1	US-01, RC-01, RC-06	#17, #18	keep	In Progress, PR #62; needs real OAuth clients. Fig. 3; state: sign-in failure
#26	Verify identity tokens at API Gateway and scope data to the signed-in student	1	RC-02	#25	keep	Agent Review, PR #62; includes the Firestore owner rules
#46	Account deletion request with confirmation	1	RC-04	#26	re-phase	In Progress, PR #62. Moved from Phase 6 (PA-11). Fig. 2 Account path
#72	Require owner approval before a signed-in account reaches the app	1	RC-01, RC-02, RC-04, PA-12	#26, #46	new	Added 2026-10-31 (POLICY). Access record, guard and Firestore rule check. Fig. 1

ID	Issue Title	Phase	SDD Sources	Depends On	Change	Notes / Visual Context
#73	Let the admin list and decide access requests with an audit record	1	RC-02, RC-06, PA-12	#72	new	Admin set server-side; versioned decisions
#74	Email the owner once for each new access request	1	RC-06, PA-12, AG-3	#72, **OD-3**	new	New data processor; name and email only
#75	Show students their access state and let them request access	1	US-01, RC-02, RC-11, PA-12	#72, #58	new	Blocks #28. No SDD figure; Fig. 2 flow
#76	Build the admin access-request queue screen	1	RC-02, RC-11, PA-12	#73, #58	new	The only admin surface (POLICY §1)
#63	Keep session content out of logs with a content-free logging helper	1	RC-03, RC-06	#18	new	Must merge before any session endpoint. #39 proves the whole flow later
#27	Define the three vetted scenario configurations with time limits and release flags	1	RC-22, DD-07, PA-5, CD-3	#18	rewrite	Persona prompts move to #67. Adds the CD-3 release flag; time limit fixed at 10 min
#22	Confirm session time limit for each scenario	—	OQ-04	—	close	Settled by CD-4 (10 min, set in config)
#24	Accessible component baseline for web and native	1	RC-11, RC-12	#17	keep	Primitives every screen uses
#57	Record approved Figma reference and screen-state-route inventory	1	RISK-05, PA-8	—	keep	Source of the Figma nodes cited in Visual Context
#58	Implement Figma visual tokens, assets, and shared presentation components	1	PA-8	#24, #57	keep	
#28	Scenario library screen	1	US-02, RC-22	#25, #27, #58, #75	keep	Hides scenarios whose release flag is off. Fig. 3
#29	Preparation screen with crisis opt-in and connectivity gate	1	US-02, RC-07, RC-13	#28	keep	Fig. 4; state: offline
#30	State the educational boundary across the product	1	RC-21	#28	keep	The report disclaimer lands with #38. Figs. 3, 4, 6
#19	Verify zero-retention and no-training terms for AI and speech providers	2	RC-05, OQ-02, DD-04	— (day one)	keep	Waits on vendors and blocks only the release. Feeds CD-1 and CD-2
#64	Define the replaceable speech provider interface and its fake	2	RC-16, DD-09	#18	new	Contract for #20 and #43; includes the contract test

ID	Issue Title	Phase	SDD Sources	Depends On	Change	Notes / Visual Context
#20	Benchmark OpenAI and Google speech and record the provider ADR	2	OQ-01, RISK-02, PA-2	#64; #19 only at the release gate (no link)	rewrite	The latency target is already set (PA-2). Measures from a device and web client (IR-3). Interface work moves to #64
#65	Configure Vertex AI for zero data retention and record the governance settings	2	RC-05, DD-04, OD-2	#18; #19 for terms status only (no link)	new	Settles OD-2 (which models). Signed vendor documents stay out of the repo
#66	Define the completed-session handoff contract	3	RC-09, RC-10, DD-02	#63	new	Phase 5 builds against it (§4 rule 3)
#31	Session service for lifecycle, persona turns, and time limits	3	US-04, RC-03, RC-09, RC-10	#63, #66, #26, #27, **OD-1**	keep	Went back to Backlog for OD-1. OD-1 was resolved on 2026-10-31 (ARCHITECTURE AD-1: signed, client-held state), so only its Issue blockers remain
#67	Drive the three vetted personas through a prompt harness on Vertex AI	3	US-04, RC-19, DD-07	#31, #65	new	Split from #27. A crisis disclosure is a turn, never a session end
#32	Text conversation screen with timer and End session	3	US-03, US-04, RC-10, RC-12	#29, #31	keep	Holds the memory-only client turn store. Fig. 5
#43	Voice mode with voice check and visible text fallback	3	US-03, DD-08	#20 (ADR), #32	keep	Stays in Backlog until the ADR merges (§4 rule 4). Fig. 5, 10; state: microphone denied
#39	Enforce session-data minimization across storage, logs, analytics, and crash reports	3	RC-03, DD-03	#32, #43	re-phase	End-to-end no-persistence proof. Adds no crash-reporting SDK (AG-3)
#48	Build the labeled conduct and crisis safeguard test set	4	RC-19, RC-20, RISK-04, PA-10(b)	#27	rewrite	Every case must pass, with no rate thresholds. Adds crisis must-not-terminate cases. Merges before #33
#33	Conduct safeguards with one warning and termination without a score	4	US-06, RC-20	#48, #31	keep	Fig. 9; states: conduct warning, conduct ended
#34	Crisis scenario in-session risk handling and post-session resources	4	RC-19	#48, #67	keep	The 988 Lifeline on the report lands with #38
#44	Recover from voice and transcription failures without scoring partial attempts	4	US-05, RC-08, RC-09	#43	keep	Includes the IR-2 state-loss test. Fig. 9; states: voice interrupted, conversation lost

ID	Issue Title	Phase	SDD Sources	Depends On	Change	Notes / Visual Context
#21	Write the rubric anchor specification from free instruments	5	RC-17, RC-18, OQ-03, PA-4	— (day one)	rewrite	Faculty review removed (PA-10). Head of the critical path
#47	Build the labeled calibration set and calibration runner	5	RISK-01, PA-10(a), CD-5	#21; #65; #35 to run	rewrite	Replaces the faculty-rating study. Lane 1, right after #21
#59	Define versioned report contract and synthetic fixtures for independent UI development	5	US-07, RC-17	—	keep	Contract for #35, #38, #40, #41 and #42
#35	Grade completed transcripts with fixed weights against the anchor specification	5	US-07, RC-09, RC-17	#21, #59, #65, #18	rewrite	Faculty-review and #47 faculty-gate references removed
#37	Crisis safety override in grading	5	RC-19, US-07	#35	keep	Every crisis critical-failure case in #47 must trigger it
#36	Validate that every feedback example is grounded in the transcript	5	RC-18, RISK-01	#35	keep	State: grading failed
#38	Feedback report screen for standard and crisis outcomes	5	US-07, RC-19, RC-21	#59, #58	keep	Figs. 6, 7, 10; score-ring percent sign fixed (RISK-05)
#42	Export the feedback report to the device and confirm before leaving	5	US-08, RC-03, PA-6	#38	keep	On-device PDF on all three platforms (IR-4). Fig. 8 left
#68	Send completed live sessions to the validated grader	5	RC-09, RC-10	#66, #31, #36, #37	new	Critical path. Only a complete transcript reaches the grader
#40	Persist scores-only records for completed sessions	6	US-09, RC-04, RC-09	#36, #68, #26, #46	keep	Server write checks the deletion marker in the same transaction. Adds completion time and crisis safety result (POLICY REP-4, REP-5)
#41	Score history screen	6	US-09, RISK-05	#40, #58	keep	Fig. 8 right; empty state fixed. Safety result first, newest first, no date (POLICY REP-4, REP-5)
#45	Verify the core flow with VoiceOver, TalkBack, and keyboard-only web	6	RC-11, RC-12	All Phase 3–6 UI	keep	Final audit (§4 rule 9)
#60	Verify Figma visual fidelity and cross-screen navigation	6	RISK-05, PA-8	All UI	keep	

ID	Issue Title	Phase	SDD Sources	Depends On	Change	Notes / Visual Context
#69	Verify cross-platform parity and produce the web and Android release builds	6	RC-14	#45	new	Covers export parity (IR-4)
#50	iOS distribution through TestFlight or internal builds	6	RC-15, DD-05	#69	keep	In Progress: Mac handoff in PR #62
#77	Provision the production GCP project for real-student use	6	AD-5, RC-05, RC-06, DD-09	#18, #65	new	Added 2026-10-31 (ROADMAP cross-check). Separate credentials; owner applies (AG-9)
#70	Review the release gates before real-student use	6	RC-05, RC-06, PA-3, PA-10	Everything above, incl. #19, #72–#76 and #77	new	The owner runs it and invokes the CD rules
#49	Faculty review of the crisis-risk scenario before release	—	RISK-03	—	close	Superseded by PA-10: #47 crisis cases, #48 crisis cases and #70

Totals across the 40 existing child Issues and the new ones: 8 new, 6 rewritten, 2 re-phased, 2 closed and 30 kept. The POLICY interview (2026-10-31) added five more new Issues, #72 to #76 (PA-12), and policy sections to #38, #40 and #41. The ROADMAP cross-check the same day added #77 for the AD-5 production project. The epics (#4, #5, #6, #11–#16, #56) stay as parents, and each new Issue joins the epic named in the backlog proposal.

6 Risks, Open Decisions & Agent Guardrails

This section adds to the SDD rather than repeating it. SDD RISK-01 to RISK-06 still apply. Below are the risks this plan's own choices create, the contingency rules decided ahead of time, the decisions still open and the limits an implementation agent must respect. `docs/POLICY.md` will expand the guardrails into the full agent policy.

6.1 Implementation risks

ID	Risk	Mitigation in this plan
IR-1	**The grader check could test itself.** Without faculty review, one author's blind spots could shape the anchors, the labels and the grader alike. A pass would then show consistency, not validity.	A model family different from the grader's drafts the calibration transcripts. The owner approves every label, and labels are frozen before grader work starts (CD-5, §4 rule 5).
IR-2	**In-memory session state on Cloud Run can vanish.** A restart, a scale-in, or a turn routed to another instance loses the conversation.	Settled by OD-1 → `docs/ARCHITECTURE.md` AD-1: the app holds the conversation with a server signature, so no instance owns it. A lost or rejected state takes the Phase 4 lost-content path, and a test simulates it. No external session store is added (AG-2).
IR-3	**Real-time audio in Expo** may need a development build with native modules, and web audio behaves differently. A server-only benchmark would hide client latency.	The Phase 2 benchmark measures from a development-build client on a device and on web.

ID	Risk	Mitigation in this plan
IR-4	**On-device PDFs differ by platform.** Fonts, page breaks and the share flow vary across web, iOS and Android.	Export is built early in Phase 5 against a fake report. The Phase 6 parity check covers export.
IR-5	**Content can leak indirectly.** The likely paths in this stack are request-body logging at the gateway or Cloud Run, error messages that echo a turn, client console logs and exception reports. RC-03 forbids all of these.	Request-body logging stays off. The logging-helper tests cover error paths. No telemetry or crash-reporting SDK is added without an owner decision (AG-3).
IR-6	**Review becomes the bottleneck.** Agents can open PRs faster than one reviewer can check them.	Two-lane WIP limit (§4 rule 1) and small Issues (§5).

6.2 Contingency rules (decided 2026-10-30)

ID	Trigger	Rule
CD-1	Written speech-vendor terms are missing at the release gate	Swap to the alternate provider through the RC-16 interface, if its written terms are on file and it meets the PA-2 latency target. If it fails either check, hold the release. Only the owner may then accept a latency miss, as a recorded exception. A terms gap is never accepted (RC-05).
CD-2	Written Vertex AI terms are missing at the release gate	Hold the release. Personas and grading both run on Vertex AI, so there is nothing to swap (OQ-02).
CD-3	The grader still fails the PA-10(a) gate after tuning	Spend one week tuning the anchor wording and the grader prompt. Labels never change. Then release only the scenarios that pass, with the failing ones flagged off in config. The 100% crisis-override rule is never relaxed. RC-22 allows a subset; this is a recorded exception to DD-07's "exactly three".
CD-4	Session time limit (PA-5, OQ-04)	10 minutes for all three scenarios, set in config.
CD-5	Who labels the calibration set (PA-10)	A model family different from the grader's drafts each transcript to a target label from the anchor spec. The owner approves every label before grader work starts.

An agent that sees a trigger reports it on the Issue and stops. The owner invokes the rule.

6.3 Open decisions

No agent may guess these. The Issues that depend on them stay blocked until the owner decides.

ID	Decision	Decide by	Blocks
OD-1	**Resolved 2026-10-31 (`docs/ARCHITECTURE.md` AD-1, AD-2).** The app holds the conversation and the session service signs it, so the services stay stateless. Rewind of a signed state is an accepted MVP risk.	Decided in the ARCHITECTURE.md interview	Nothing (it blocked the Phase 3 session endpoints)
OD-2	Which Vertex AI models run the personas and the grader, and so which family drafts the calibration set	Phase 2 data-governance record	Persona harness, grader, calibration set drafting
OD-3	Which email provider sends the owner's access-request emails, with its data-use record (it receives only the requester's name and Google email; AG-3)	Before #74 starts	#74 owner notification email

6.4 Agent guardrails

1. **Requirements win.** If an Issue conflicts with an RC, a PA or an SDD non-goal, stop and ask. Never settle the conflict in code.
2. **No session content at rest.** Audio, transcripts, excerpts and narrative feedback never go to Firestore, logs, analytics, crash reports, caches, an external session store or test fixtures. All test data is synthetic (RC-03, PA-3).
3. **No new data processors.** Adding a third-party service, SDK or model API takes an owner decision and a zero-retention record first (RC-05). This includes telemetry and crash reporting.
4. **Grading integrity.** Code computes the overall score from the fixed RC-17 weights. Never change the weights or the crisis override. Never edit calibration labels or safeguard test cases to make a test pass.
5. **Licensed instruments.** Never reproduce CCS-R or MITI manual text, and never label a score as an MITI score (PA-4, RC-18).
6. **Fixed product scope.** Google sign-in only. No live scores or hints during a session. No scenario, persona or rubric editing. No instructor features (RC-01, RC-10, RC-22, SDD §2.3).
7. **Security defaults.** Secrets come only from Secret Manager, and none reach the client. Every endpoint checks the caller's identity at the gateway and again in the service (RC-02, RC-06).
8. **Prototype is a reference.** Build from the figures and states. Never copy Figma Make markup (PA-8).
9. **Humans own releases.** No deploys, no merges to `staging` or `main` and no real-student enablement. The owner runs the release gate review and invokes the CD rules.

7 Handoff to GitHub Project

The backlog lives on Project #1: 10 epics, 46 child Issues and 90 dependency links, organized on 2026-10-31. The POLICY interview later that day added 5 child Issues (#72 to #76, Human Review Pending) and 14 links, for 51 and 104. The ROADMAP cross-check added #77 (production project) and 3 links, for 52 and 107, and moved four pre-Lab 3 Sequence values into phase order (#43, #39, #44, #42). The final consistency pass the same day brought 11 Issue bodies in line with PA-6, PA-10 and PA-12 (#4, #16, #25, #34, #38, #42, #56 to #60), with a comment on each; no Status or link changed. From that date, **GitHub Issues and Project #1 are the authoritative source for detailed work**: what an Issue requires, its status, sequence, dependencies and review state. This plan stays the higher-level explanation of why the work is phased and ordered as it is. If an Issue and this plan disagree, the agent stops and asks (AG-1). The owner then fixes the Issue for a task detail, or this plan for phase intent.

How the plan maps to the Project

Plan concept	Project field or feature	Values
Plan phase (§2)	**Plan Phase**	1 Foundation & access · 2 Provider & compliance spike · 3 Practice session · 4 Safety & recovery · 5 Grading & report · 6 History & release. Epics have none, because they span phases
SDLC activity	**Stage** (named "Phase" until 2026-10-31; renamed so it can't be confused with Plan Phase)	Planning, Design, Implementation, Testing, Deployment
Order (§4.1)	**Sequence**	Board order. Decimals slot new work between existing Issues

Plan concept	Project field or feature	Values
Dependencies (§4.1, §5.1 "Depends On")	Native **blocked by** links	107 links (90 at handoff, then 14 for #72 to #76 and 3 for #77). #19 (vendor terms) is linked only to #70, because terms block the release, not development (§4)
Open decisions (§6.3)	Status stays Backlog, with a comment naming the OD	#31 waited on OD-1 until 2026-10-31
Review state	**Human Review**	Pending → Reviewed
Review depth (§5)	**Review Depth**	Owner decision · Deep review · Standard review
Epics (§5 sizing rule 6)	Parent issue and **Sub-issues progress**	#4, #5, #6, #11–#16, #56
Ownership and kind of work	**Priority**, **Area**	Unchanged from the backlog proposal
Two lanes (§4 rule 1)	Count of child Issues In Progress	At most one critical-path and one off-path Issue

Status rules

- **Backlog:** a blocker is still open, meaning a blocked-by Issue is neither merged nor in an open PR, or an open decision applies.
- **Ready:** every blocker is merged or in an open PR, and no open decision applies. On 2026-10-31 that was 10 child Issues: #19, #21, #24, #27, #51, #57, #59, #63, #64 and #65. Twenty-one child Issues and three epics moved from Ready to Backlog. #72 joined them later that day, after the POLICY interview, for 11.
- **In Progress onward:** as in the project `CLAUDE.md`. The seven Issues in PR #62 (#17, #18, #25, #26, #46, #50, #61) kept their status.
- **Epics** follow their children. An epic is In Progress if any child is in flight, Ready if any child is Ready, and Backlog otherwise.
- **Promotion:** when a blocker merges, the next agent session moves the Issues it unblocked from Backlog to Ready and posts a comment saying which merge unblocked them.

Human Review gate

1. Every open epic and child Issue starts at **Pending**: 56 items on 2026-10-31, and 62 once #72 to #77 were added the same day. #7 (Lab 2 process work, Released) and the closed #22 and #49 have no value.
2. The owner sets **Reviewed** only after checking the Issue's scope, requirements, acceptance criteria, visual references, dependencies, testing expectations and Project fields.
3. **An agent starts an Issue only when Status is Ready and Human Review is Reviewed.** It may read and refine a Pending Issue, but not implement it.
4. When an agent changes an Issue's scope or acceptance criteria, it resets Human Review to Pending and says why in a comment.
5. Work already in flight continues. PR #62 merges only after its Issues are Reviewed.

Suggested first reviews, one per depth and all Ready: **#21** (Owner decision, head of the critical path), **#63** (Deep review, which every session endpoint needs) and **#57** (Standard review, the source of the Figma nodes in every UI Issue).

Traceability

Each step below can be followed on GitHub through the Issue body's SDD IDs, the Plan Phase field and the blocked-by links.

- **US-07 / RC-17** (weighted, evidence-linked feedback) → Phase 5 (§3.5) → #21 anchor spec → #47 calibration set → #35 grader → #36 evidence validation → #68 live integration.
- **RC-03** (no session content at rest) → Phase 1 (§3.1) → #63 content-free logging helper → Phase 3 #39 no-persistence proof.

Known gaps at handoff

- **Issue bodies.** The 14 new or rewritten Issues follow the §5 structure. The 32 kept or re-phased Issues predate it. They have acceptance criteria, but most lack Dependencies and Visual Context sections. Their dependencies are now native links, and their visual context is in the §5.1 Notes column. The owner chose links over body rewrites (2026-10-31). The Human Review pass is where an Issue gets flagged if that isn't enough.
- **Plan Phase view.** GitHub's API can create a Project view but cannot set its grouping. A view grouped by Plan Phase is a one-minute owner action in the GitHub UI.
- **OD-1** was resolved in the ARCHITECTURE.md interview on 2026-10-31 (AD-1). #31 and the Phase 3 chain now wait only on their Issue blockers.
- **#50** is In Progress, but only for the Mac handoff in PR #62. The TestFlight build itself is blocked by #69.

7.1 Final Readiness Checklist

Check	Result	Evidence
Every MVP user story is represented by at least one phase and Issue	Met	US-01 #25 · US-02 #28, #29 · US-03 #32, #43 · US-04 #31, #67 · US-05 #44 · US-06 #33 · US-07 #35, #37, #38 · US-08 #42 · US-09 #40, #41
Every phase has concrete deliverables and objective acceptance criteria	Met	§3.1–§3.6 Deliverables and Phase Acceptance Criteria
Dependencies and sequencing are explicit enough that an agent does not guess the order of high-risk work	Met	§4 rules, §4.1 table, 107 blocked-by links, and Ready only when unblocked
Relevant UI/UX and architecture references are identified for future Issues	Met in the plan; partly met in Issue bodies	§3 "Relevant UI/UX Artifacts" and §5.1 Notes; Visual Context sections exist only in the 14 new or rewritten Issues (see Known gaps)
The Issue inventory is detailed enough to create Issues without redesigning the project	Met	All 52 child Issues exist on Project #1 (§5.1)
SDD, plan, Issues and the three docs agree	Met	Final consistency pass, 2026-10-31: every SDD override is a recorded PA (PA-4, PA-6, PA-10, PA-12) and no Issue body still contradicts one
Every implementation-ready Issue is human-reviewed	**Not yet**	All 62 items are Pending. The owner's review is the last step before agents start (gate rule 3)